

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

**ENVIRONMENTAL SCIENCE****Paper 2 (Generated practice paper - Set 37)**

Assessed: The living environment, Biological resources, Sustainability, Research methods

**Time allowed: 3 hours****Materials**

For this paper you may use:

- a calculator.

**Instructions**

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

**Information**

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

**For Examiner's Use**

| Question | Mark |
|----------|------|
| 1        |      |
| 2        |      |
| 3        |      |
| 4        |      |
| 5        |      |
| 6        |      |
| 7        |      |
| 8        |      |
| 9        |      |
| 10       |      |
| 11       |      |
| TOTAL    |      |

**Answer all questions in the spaces provided.**

**0 1 . 1**

Describe coppicing and explain how it increases the biodiversity of a woodland.

**[4 marks]**

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**0 1 . 2**

Explain how the reintroduction of grey wolves to Yellowstone National Park increased biodiversity.

**[4 marks]**

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**0 1 . 3**

Explain why culling or removing species may be necessary in habitat management.

**[2 marks]**

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10

0 2 . 1 Explain the effects of deforestation on soil.

[4 marks]

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0 2 . 2 Explain how deforestation affects hydrology and climate.

[4 marks]

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0 2 . 3 Give **two** ecological impacts of deforestation other than the direct loss of species.

[2 marks]

1 .....

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2 .....

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10

03 . 1

Explain why introduced species often have a greater effect on island communities than on mainland communities.

[2 marks]

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03 . 2

For each of the following, describe an example of an introduced species and explain its effect on an indigenous species.

[6 marks]

**Introduced competitor** .....

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**Introduced predator** .....

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**Introduced pathogen** .....

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03 . 3

Explain how hybridisation with an introduced species threatens the red deer and the Scottish wildcat.

[2 marks]

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0 4

Table 1 shows climate data for two sites where a farmer is considering growing maize, a frost-sensitive crop that needs a thermal growing season of at least 120 days.

Table 1

| Feature                               | Site A: valley floor | Site B: south-facing slope |
|---------------------------------------|----------------------|----------------------------|
| Frost-free period / days              | 110                  | 150                        |
| Days with mean temperature above 5 °C | 160                  | 175                        |
| Mean summer rainfall / mm             | 180                  | 150                        |
| Gradient / %                          | 1                    | 8                          |

0 4

. 1

Use Table 1 to explain which site is more suitable for maize.

[3 marks]

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0 4

. 2

Describe **three** ways in which temperature can be controlled to increase crop productivity.

[3 marks]

1 .....

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2 .....

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3 .....

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0 4

. 3

Explain why keeping livestock warm increases the amount of food converted to growth.

[2 marks]

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0 4 . 4

Explain how day length affects livestock production and how it can be manipulated.

[2 marks]

10

0 5

Table 1 compares extensive oyster farming with intensive salmon farming.

Table 1

| Feature                | Extensive oyster farm                          | Intensive salmon farm                                     |
|------------------------|------------------------------------------------|-----------------------------------------------------------|
| Food input             | None - filter natural plankton                 | Fishmeal / plant pellets                                  |
| Stocking density       | Low, spaced on seabed                          | High, in sea cages / tanks                                |
| Breeding               | Wild spat settle on cleared seabed or hatchery | Selected adults stripped of eggs and milt                 |
| Disease / pest control | Removal of predatory crabs                     | Antibiotics, pesticides, wrasse, removal of diseased fish |
| Yield per hectare      | Low                                            | Very high                                                 |

0 5 . 1

Use Table 1 to explain the relative merits of extensive and intensive aquaculture.

[4 marks]

0 5 . 2

Describe the stages in breeding and rearing salmon in an intensive farm.

[3 marks]

0 5 . 3

Explain how shrimp farming has affected mangroves and how disease is controlled in shrimp lagoons.

[3 marks]

10

0 6 . 1

Give **two** natural processes that are driven by solar energy.

[2 marks]

1 .....

2 .....

0 6 . 2

Explain why natural processes can take place at low temperatures.

[2 marks]

06 . 3

Give **one** human activity that uses high temperatures to achieve a result that natural systems achieve at low temperature.

[1 mark]

5

07

A student estimated the population of ground beetles in a field using the Lincoln Index. 60 beetles were caught in pitfall traps, marked with a spot of paint and released. Three days later 45 beetles were caught, of which 9 were marked.

07 . 1

Calculate the estimated population size.

[2 marks]

Answer

beetles

07 . 2

State **four** assumptions of the Lincoln Index method.

[4 marks]

1 .....

2 .....

3 .....

4 .....

07 . 3

Explain how the estimate would be affected if the paint made the beetles more likely to be eaten by birds.

[2 marks]

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07 . 4

Suggest why the second sample was taken after three days rather than the next day or after a month.

[2 marks]

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10

08 . 1

Explain how each of the following new technologies improves the validity of ecological research.

[8 marks]

**Satellite / radio / GPS tracking** .....

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**DNA databases and eDNA** .....

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**Image recognition software** .....

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**Acoustic monitoring / sonograms** .....

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**08 . 2**

Explain why new technologies allow the collection of 'more representative' data.

**[2 marks]**

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**10**

09

Table 1 shows the numbers of breeding pairs of two seabird species on an island before and after a rat eradication programme.

Table 1

| Species         | Breeding pairs before eradication | Breeding pairs 8 years after eradication |
|-----------------|-----------------------------------|------------------------------------------|
| Manx shearwater | 45                                | 310                                      |
| Storm petrel    | 12                                | 140                                      |

09

1

Explain why oceanic islands have many endemic species but low overall biodiversity.

[4 marks]

09

2

Use Table 1 to explain why introduced rats had such a severe effect on these birds.

[3 marks]

09

3

Explain how avian malaria threatens Hawaiian birds and how climate change worsens the threat.

[2 marks]

09 . 4

Give **one** other conservation measure for oceanic islands besides eradicating introduced species.

[1 mark]

10

10 . 1

Explain how an understanding of ecological succession and population dynamics can inform conservation decision-making.

[9 marks]

[illegible]

**[1 mark]**

Do not write outside the box

10

**Write an essay on one of the following topics.**

$$\boxed{1}\boxed{1}.\boxed{1}$$

Using tropical coral reefs as an example, discuss how the ecological features of a habitat determine both its importance and its vulnerability to human activities.

**[25 marks]**

**OR**

1 1 . 2

Discuss the causes and consequences of deforestation and evaluate the extent to which forests can be exploited sustainably.

**[25 marks]**

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.



[illegible]

[illegible]

Do not write outside the box

**END OF QUESTIONS**